

Algorithm 6: Library Management System

Exercise 6: Library Management System

Scenario:

You are developing a library management system where users can search for books by title or author.

Steps:

Understand Search Algorithms:

Explain linear search and binary search algorithms.

Setup:

Create a class Book with attributes like bookId, title, and author.

Implementation:

Implement linear search to find books by title.

Implement binary search to find books by title (assuming the list is sorted).

Analysis:

Compare the time complexity of linear and binary search.

Discuss when to use each algorithm based on the data set size and order.